

**ENGINEERING
ADMISSIONS ASSESSMENT****D564/32****November 2020****60 minutes****SECTION 2****INSTRUCTIONS TO CANDIDATES**

Please read these instructions carefully, but do not open this question paper until you are told that you may do so. This paper is Section 2 of 2.

A separate answer sheet is provided for this paper. Please check you have one. You also require a soft pencil and an eraser.

Please complete the answer sheet with your candidate number, centre number, date of birth, and name.

This paper contains 20 multiple-choice questions. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt **all** 20 questions. Each question is worth one mark.

For each question, choose the **one** option you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

You **must** complete the answer sheet within the time limit.

You can use the question paper for rough working, but **no extra paper** is allowed. Only your responses on the answer sheet will be marked.

Dictionaries and calculators are NOT permitted.

Please wait to be told you may begin before turning this page.

This question paper consists of 22 printed pages and 2 blank pages.



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- 1** Spring P has spring constant 1.0Ncm^{-1} and spring Q has spring constant 3.0Ncm^{-1} .

The two springs are connected in series.

The springs are stretched by 6.0cm in total.

What is the extension of spring P?

(The springs have negligible mass and obey Hooke's law.)

- A** 1.5cm
- B** 2.0cm
- C** 3.0cm
- D** 4.0cm
- E** 4.5cm

- 2 A single strand of wire has a radius of $2.0 \times 10^{-4} \text{ m}$ and length 15 m. The resistivity of the material from which the wire is made is $4.8 \times 10^{-7} \Omega \text{ m}$.

Twelve strands of this wire are connected in parallel to make a cable.

What is the resistance of the cable?

A $\frac{\pi}{2160} \Omega$

B $\frac{\pi}{180} \Omega$

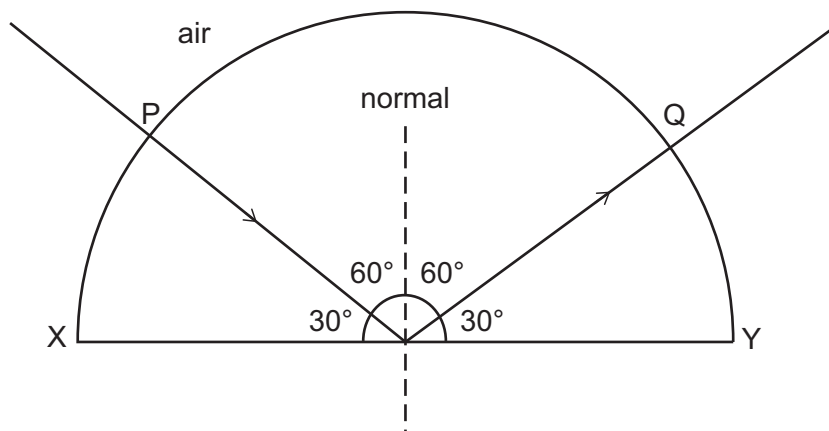
C $\frac{\pi}{15} \Omega$

D $\frac{15}{\pi} \Omega$

E $\frac{180}{\pi} \Omega$

F $\frac{2160}{\pi} \Omega$

- 3 A ray of light is directed into a semicircular transparent block, entering at P. The direction of the ray is adjusted until it strikes the centre of the flat face XY of the block at the critical angle and reflects to Q as shown.



The length of XY is L .

The speed of light in air is c .

What is the time taken by the light to travel from P to Q in the block?

- A $\frac{L\sqrt{3}}{2c}$
- B $\frac{L}{c}$
- C $\frac{2L}{c\sqrt{3}}$
- D $\frac{L\sqrt{3}}{c}$
- E $\frac{2L}{c}$
- F $\frac{4L}{c\sqrt{3}}$

- 4 A solid cube with sides of length 20 cm is made from material with density 2000 kg m^{-3} . The cube is suspended, in equilibrium, from an initially unstretched spring, and this results in the spring gaining strain energy of 3.2 J.

What is the spring constant of the spring?

(gravitational field strength = 10 N kg^{-1} ; the spring obeys Hooke's law)

- A 40 N m^{-1}
- B 80 N m^{-1}
- C 400 N m^{-1}
- D 800 N m^{-1}
- E 4000 N m^{-1}
- F 8000 N m^{-1}

- 5 A projectile is fired upwards from the ground at an angle of 60° to the vertical at a speed of 20 m s^{-1} .

It travels a horizontal distance d and lands with a downwards vertical component of velocity of 4.0 m s^{-1} on ground that is height h above the starting point of the projectile.

What are d and h ?

(gravitational field strength = 10 N kg^{-1} ; assume that air resistance is negligible)

	d / m	h / m
A	$6.0\sqrt{3}$	4.2
B	$6.0\sqrt{3}$	5.8
C	$10\sqrt{3} - 4.0$	4.2
D	$10\sqrt{3} - 4.0$	14.2
E	$10\sqrt{3} + 4.0$	5.8
F	$10\sqrt{3} + 4.0$	14.2
G	$14\sqrt{3}$	4.2
H	$14\sqrt{3}$	5.8

- 6 Diagram 1 shows the positions of nine equally spaced particles in a medium.

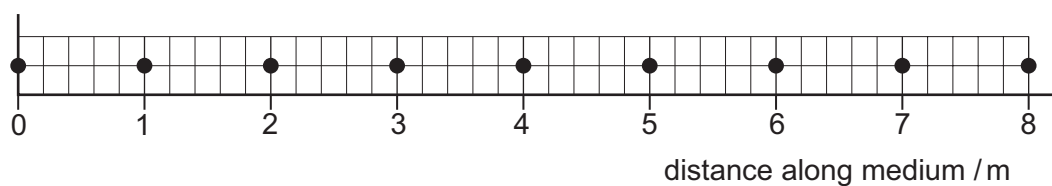


Diagram 1

Diagram 2 shows the positions of the same nine particles, at a particular time, while a longitudinal wave is travelling through the medium.

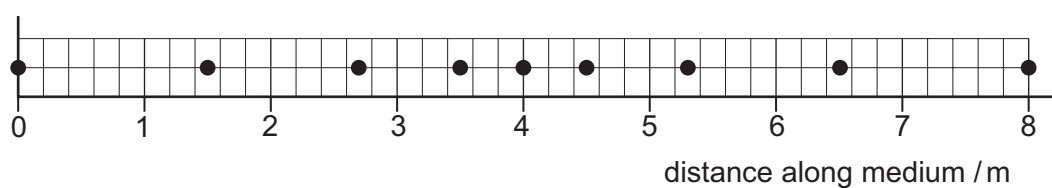


Diagram 2

What is the amplitude of the wave?

- A 0.4 m
- B 0.5 m
- C 0.6 m
- D 0.7 m
- E 2.0 m
- F 4.0 m
- G 6.0 m
- H 8.0 m

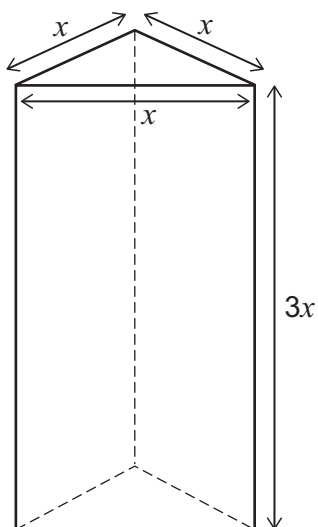
- 7 A spaceship with mass $8.0 \times 10^4 \text{ kg}$ travels at constant velocity and has $1.0 \times 10^{12} \text{ J}$ of kinetic energy.

An external impulse of $8.0 \times 10^7 \text{ kg m s}^{-1}$, lasting for 2.0 s, is applied to the spaceship acting in the opposite direction to the motion of the spaceship.

What is the average rate of loss of kinetic energy of the spaceship during the application of the impulse?

- A $9.5 \times 10^{10} \text{ W}$
- B $1.8 \times 10^{11} \text{ W}$
- C $2.2 \times 10^{11} \text{ W}$
- D $3.2 \times 10^{11} \text{ W}$
- E $3.6 \times 10^{11} \text{ W}$
- F $7.2 \times 10^{11} \text{ W}$

- 8 The diagram shows a solid triangular prism.



The sides of the triangular cross section of the prism are of length x .

The height of the prism is $3x$.

The uniform density of the prism is ρ .

The gravitational field strength is g .

What is the minimum pressure the prism can exert when it rests on level ground?

- A $3\rho g$
- B $3\rho gx$
- C $\frac{\rho g}{4}$
- D $\frac{\rho gx}{4}$
- E $\frac{\sqrt{3}\rho g}{4}$
- F $\frac{\sqrt{3}\rho gx}{4}$

- 9 An apple of mass m_a is placed on a uniform metre rule with the centre of gravity of the apple at the 10 cm mark. The rule is balanced on a pivot placed at the 35 cm mark.

The apple is replaced with an orange of mass m_o . The rule now balances with the pivot at the 40 cm mark.

What is the ratio $\frac{m_a}{m_o}$?

A $\frac{5}{9}$

B $\frac{4}{5}$

C $\frac{5}{6}$

D $\frac{6}{5}$

E $\frac{5}{4}$

F $\frac{9}{5}$

- 10** A cyclist travels at a constant speed of 12 m s^{-1} on level ground. During this time the power needed to maintain a constant speed is 900 W . The total weight of the cyclist and bicycle is 850 N .

The cyclist now cycles up a slope at the same constant speed. The slope is at an angle of 30° to the horizontal.

What is the driving force on the bicycle as it travels up the slope?

(Assume that the magnitude of the resistive forces is constant.)

- A** 75 N
- B** 350 N
- C** 500 N
- D** $(425\sqrt{3} - 75) \text{ N}$
- E** 775 N
- F** $(425\sqrt{3} + 75) \text{ N}$
- G** 925 N

- 11** Three identical resistors can be combined in four different arrangements.

One of the arrangements has a resistance of $18\,\Omega$.

A different arrangement has a resistance of $8.0\,\Omega$.

What are the resistances of the other two arrangements?

(All three resistors contribute to the total resistance in all arrangements.)

A $2.0\,\Omega$ and $4.0\,\Omega$

B $2.0\,\Omega$ and $9.0\,\Omega$

C $4.0\,\Omega$ and $12\,\Omega$

D $4.0\,\Omega$ and $36\,\Omega$

E $36\,\Omega$ and $162\,\Omega$

F $81\,\Omega$ and $162\,\Omega$

- 12** A $4.0\text{ k}\Omega$ fixed resistor is connected in series with a light dependent resistor (LDR) across a 100 V dc power supply.

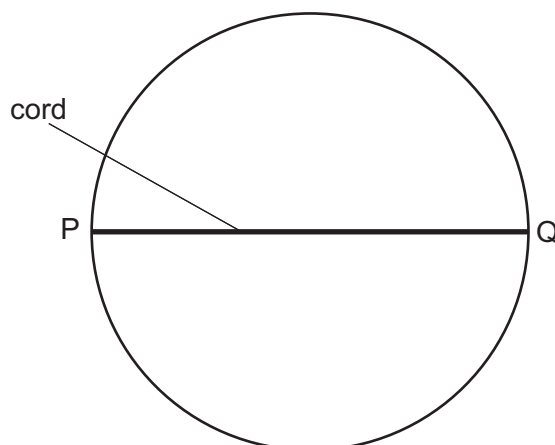
The current in the LDR is 5.0 mA .

The intensity of light falling on the LDR now decreases and the voltage across the fixed resistor changes by 50% .

What is the change in the resistance of the LDR as a result of the change in intensity?

- A** $8.0\text{ k}\Omega$
- B** $12\text{ k}\Omega$
- C** $16\text{ k}\Omega$
- D** $20\text{ k}\Omega$
- E** $32\text{ k}\Omega$
- F** $36\text{ k}\Omega$

- 13 An elastic cord with spring constant k is fixed to two points P and Q on the diameter of a ring so that the cord is taut but unstretched. The radius of the ring is r .



The midpoint of the cord is then pulled and fixed to a point on the ring halfway between P and Q.

What is the energy stored in the elastic cord?

- A $\frac{1}{2}kr^2$
- B $2kr^2$
- C $\frac{1}{2}(\sqrt{2} - 1)kr^2$
- D $2(\sqrt{2} - 1)kr^2$
- E $\frac{1}{2}(3 - 2\sqrt{2})kr^2$
- F $2(3 - 2\sqrt{2})kr^2$

- 14** An object of mass M experiences a resultant force of magnitude F . The force acts in a single horizontal direction with a magnitude that varies with time t according to

$$F = X + Y\sqrt{t}$$

where X and Y are constants.

The object is at rest at $t = 0$.

What is the magnitude of the momentum of the object at time $t = T$?

- A** $T\left(X + \frac{2}{3}Y\sqrt{T}\right)$
- B** $T\left(X + Y\sqrt{T}\right)$
- C** $\frac{T}{M}\left(X + \frac{2}{3}Y\sqrt{T}\right)$
- D** $\frac{T}{M}\left(X + Y\sqrt{T}\right)$
- E** $\frac{Y}{2\sqrt{T}}$
- F** $\frac{Y}{2M\sqrt{T}}$

- 15** A trolley of mass 3.0 kg is moving horizontally along a smooth track. Its displacement x from a point at time t is given by the equation:

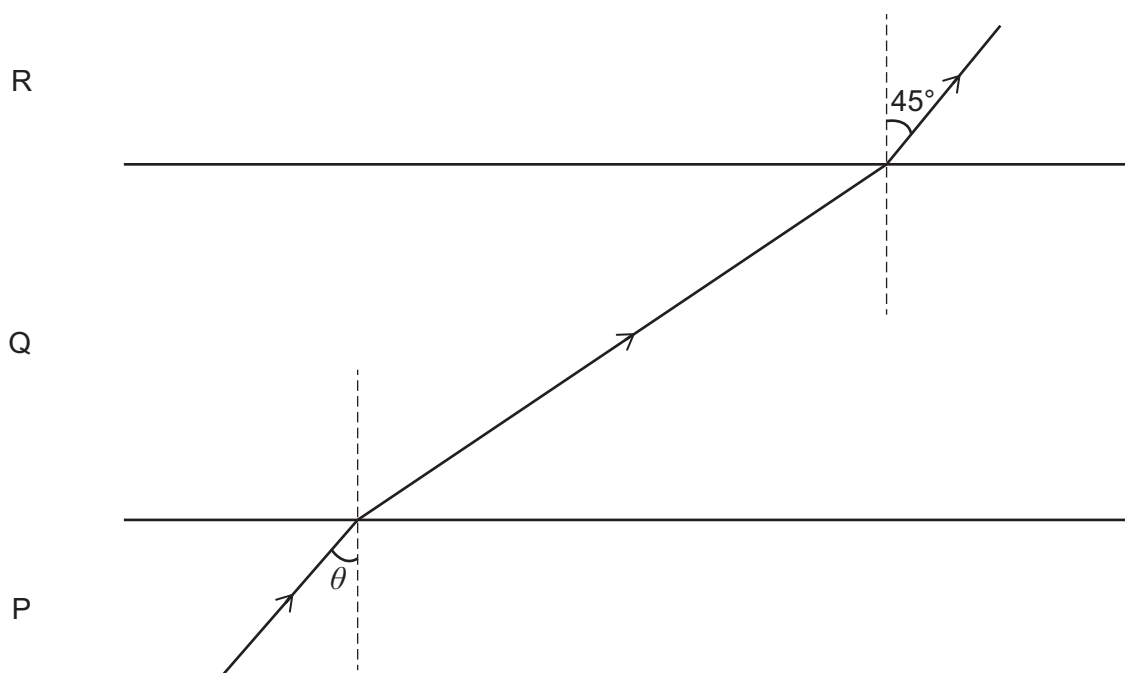
$$x = 8 + 4t + 2t^2$$

where x is in metres and t is in seconds.

How much work is done on the trolley between times $t = 0$ and $t = 5.0$ s?

- A** 12 J
- B** 24 J
- C** 78 J
- D** 270 J
- E** 840 J
- F** 864 J
- G** 936 J

- 16 The diagram shows a ray of light passing through three mediums, P, Q and R. The boundaries between the three mediums are parallel.



[diagram not to scale]

The ratio of the speed of light in medium P to the speed of light in medium Q is $2 : \sqrt{5}$

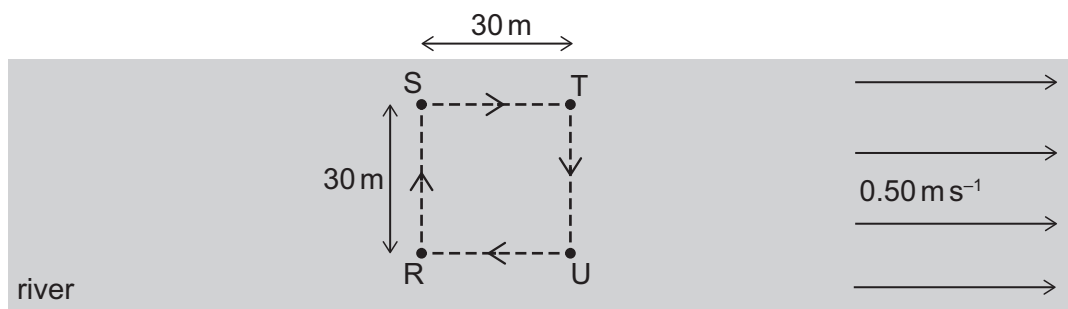
The ratio of the speed of light in medium Q to the speed of light in medium R is $3 : \sqrt{6}$

What is the value of $\sin \theta$?

- A $\frac{\sqrt{2}}{2}$
- B $\frac{\sqrt{3}}{2}$
- C $\frac{\sqrt{3}}{6}$
- D $\frac{\sqrt{5}}{5}$
- E $\frac{\sqrt{15}}{5}$
- F $\frac{\sqrt{15}}{6}$

- 17 Water in a wide river flows at a constant speed of 0.50 m s^{-1} . A swimmer swims around a square path of side 30 m marked out by 4 posts R, S, T and U which are fixed to the river bed, as shown.

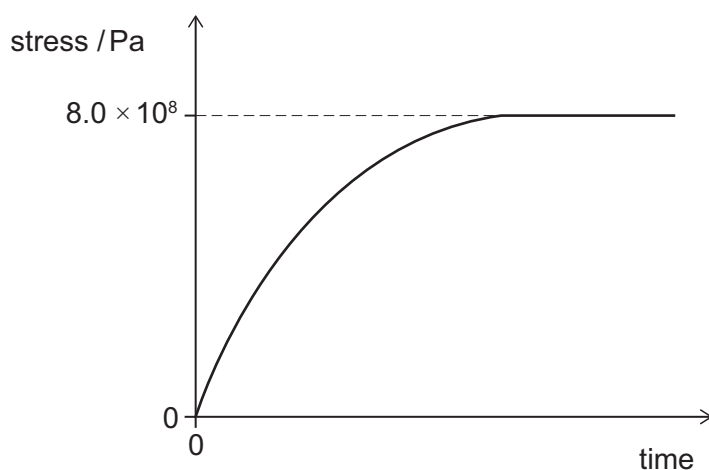
The swimmer has a constant speed of 1.0 m s^{-1} relative to the water.



How long does it take for the swimmer to swim around the square path once?

- A $(60 + 24\sqrt{5}) \text{ s}$
- B $(60 + 40\sqrt{3}) \text{ s}$
- C $(80 + 24\sqrt{5}) \text{ s}$
- D $(80 + 40\sqrt{3}) \text{ s}$
- E 120 s
- F 140 s

- 18** The stress in a steel cable increases with time and is then maintained at a constant value, as shown. The wire does not reach its limit of proportionality.



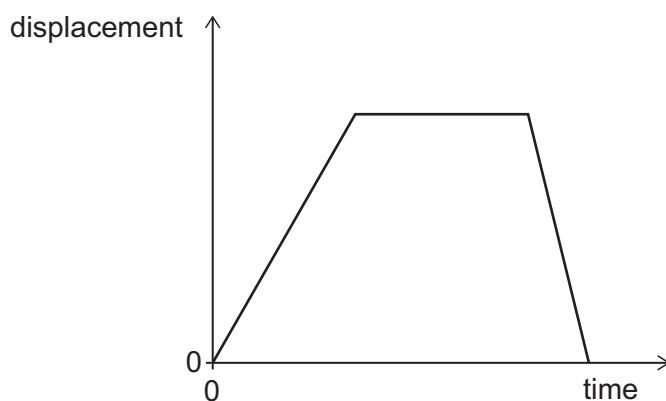
The table shows properties of the steel used in the cable and the dimensions of the cable.

<i>length / m</i>	<i>cross-sectional area / m²</i>	<i>Young modulus / Pa</i>
4.0	2.0×10^{-4}	2.0×10^{11}

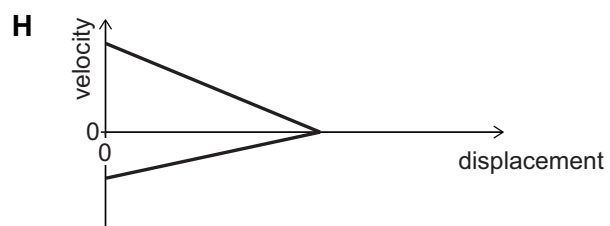
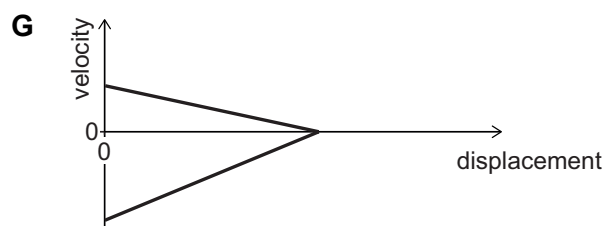
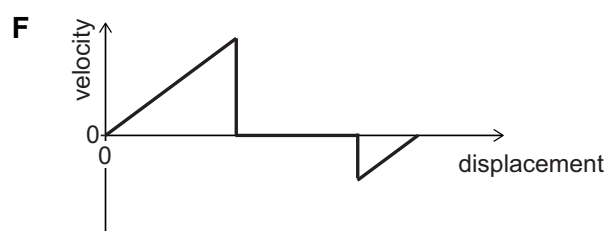
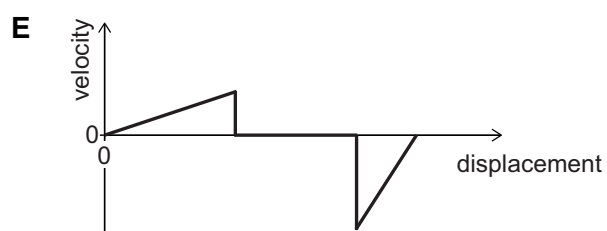
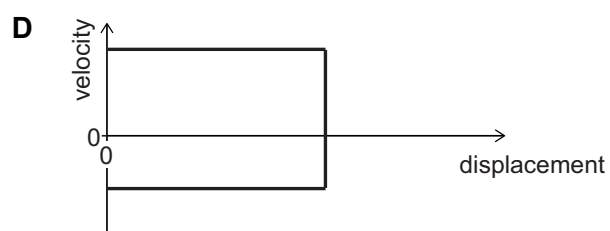
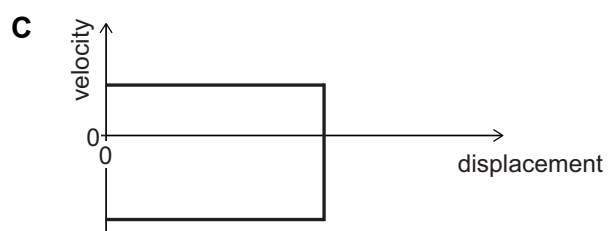
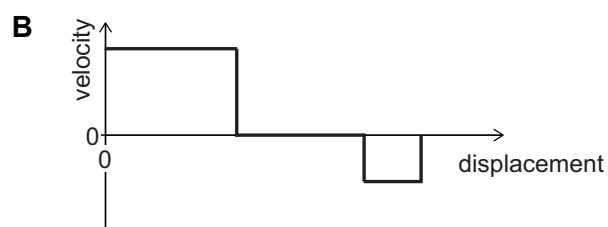
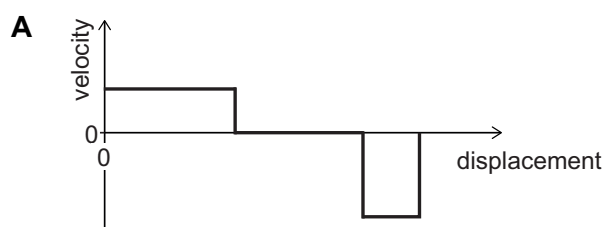
How much work was done to stretch the cable?

- A** 320 J
- B** 1.28 kJ
- C** 2.56 kJ
- D** 320 kJ
- E** 640 kJ
- F** 1.60 MJ
- G** 6.40 MJ

- 19** The following graph shows how the displacement of an object travelling along a straight, horizontal track varies with time.



Which graph shows the velocity of this object against displacement?



- 20** A cell has emf E and internal resistance r that varies with current I according to:

$$r = kI^2$$

where k is a constant.

A variable resistor is connected to the terminals of the cell. The resistance of the variable resistor is adjusted.

Which expression gives the resistance of the variable resistor, in terms of k and E , that causes maximum power dissipation in it?

A $3\left(\frac{kE^2}{2}\right)^{\frac{1}{3}}$

B $3\left(\frac{kE^2}{4}\right)^{\frac{1}{3}}$

C $3\left(\frac{kE^2}{9}\right)^{\frac{1}{3}}$

D $3\left(\frac{kE^2}{16}\right)^{\frac{1}{3}}$

E $\left(2kE^2\right)^{\frac{1}{3}}$

F $\left(4kE^2\right)^{\frac{1}{3}}$

G $\left(9kE^2\right)^{\frac{1}{3}}$

H $\left(16kE^2\right)^{\frac{1}{3}}$

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